Exploring Data: Web Scraping, Data Analysis, and Dashboard Creation:REPORT

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Project:
Individual Project Report: Web Scraping and Exploratory Data Analysis of Books to Scrape

Individual Work: Web Scraping and Exploratory Data Analysis of Books to Scrape

1. Introduction

1.1 Project Overview

This individual project focused on web scraping, data cleaning, and exploratory data analysis (EDA) using Python. The objective was to extract structured information from an online book catalogue, prepare the collected data for analysis, and identify patterns relating to book prices, ratings, and availability.

The data was collected from Books to Scrape, a demonstration website designed for practising web scraping. The project used Python libraries, including Requests, BeautifulSoup, Pandas, NumPy, and Matplotlib, to retrieve, process, analyse, and visualize the data.

1.2 What Data Was Scraped?

The project collected **1,000 book records across 50 pages** of the website. Seven columns were captured for each book:

- **Page:** The catalogue page from which the book was collected.
- **Title:** The title of the book.
- **Price (£):** The listed price in British pounds.
- **Rating:** The book's star rating, ranging from one to five stars.
- **Availability:** The displayed stock availability status.
- **Book URL:** The direct link to the book's product page.
- **Image URL:** The link to the book's cover image.

1.3 Why Was This Dataset Chosen?

The dataset was selected because it provided an opportunity to practise the complete web scraping workflow using publicly accessible catalogue information. The website also had a clear, repeated structure, making it suitable for learning how to locate HTML elements, extract information, and navigate multiple pages.

Furthermore, the dataset contained both categorical and numerical variables. This made it useful for practising data cleaning, descriptive statistics, correlation analysis, grouping, and visualization.

The project also provided practical experience in converting unstructured website content into a structured dataset that could be exported to CSV and Excel formats for further analysis.

2. Data Cleaning

2.1 Data Extraction and Preparation

The data preparation process began with importing the required Python libraries. The Requests library was used to retrieve the website's HTML content, while BeautifulSoup was used to parse the HTML and locate individual book records.

The initial website request returned an HTTP status code of 200, indicating a successful response. The page structure was inspected, and the book listings were identified using their HTML elements.

A custom scraping function was developed to extract the seven required fields from each book listing. The function was then used in a loop to navigate through the catalogue's pagination links until all available pages had been processed.

The extracted records were stored in a Python list and subsequently converted into a Pandas DataFrame for cleaning and analysis.

2.2 Cleaning and Transformation Steps

1. Price conversion

The book prices were extracted as text containing the pound currency symbol. These values were cleaned by removing the currency symbol and any unwanted character associated with text encoding before converting the prices to floating-point numbers.

This transformation made it possible to calculate the mean price, identify the cheapest and most expensive books, and examine price distributions.

2. Rating conversion

The website represented ratings as words, such as *One*, *Two*, *Three*, *Four*, and *Five*. These values were converted into numerical scores using a Python dictionary:

- One = 1
- Two = 2
- Three = 3
- Four = 4
- Five = 5

Converting ratings into numbers made them suitable for frequency analysis, grouping, and correlation calculations.

3. Availability text extraction

The availability information was extracted from the relevant HTML element using BeautifulSoup's text extraction method. Whitespace was removed, and the text was normalized into a readable format.

This enabled the availability field to be grouped and summarized consistently.

4. URL handling

The book and image links were processed using Python's `urljoin()` function. This converted relative links into complete URLs, allowing the dataset to retain direct links to individual product pages and book images.

5. Data structuring

The extracted information was organized into a Pandas DataFrame with clearly defined column names. The resulting dataset contained 1,000 rows and seven columns.

The cleaned dataset was then exported to CSV and Excel files for reuse outside the notebook.

2.3 Challenges Encountered

Several technical challenges were addressed during the scraping and preparation process.

Pagination: The catalogue was spread across multiple pages. Scraping only the first page would have produced an incomplete dataset. This was addressed by identifying the website's next-page link and using a loop to continue scraping until no further page was available.

Data type conversion: Prices and ratings were initially represented as text. These fields required conversion into numerical formats before statistical analysis could be performed.

Currency and text encoding: The price strings required cleaning to remove the currency symbol and an unwanted encoding character that could interfere with conversion to numerical values.

Relative URLs: Some links were relative to the current page rather than complete web addresses. The `urljoin()` function was used to construct complete URLs.

Responsible request handling: A one-second pause was included between page requests to reduce the frequency of requests sent to the website.

2.4 Data Quality Considerations

Although the notebook successfully extracted and transformed the dataset, additional validation would strengthen the cleaning process. In particular, checking for missing values, duplicate records, invalid prices, and unexpected rating values would help confirm the dataset's overall quality.

These checks were not explicitly documented in the current notebook and should be performed before treating the dataset as fully validated.

3. Exploratory Data Analysis (EDA)

3.1 Objective of the Analysis

The EDA stage aimed to understand the distribution of book prices and ratings, identify the cheapest and most expensive books, examine stock availability, and investigate whether book ratings were associated with their prices.

Pandas was used to calculate summary statistics and grouped results, while Matplotlib was used to produce charts for interpreting the data.

3.2 Key Findings

1. Dataset size and structure

The final DataFrame contained 1,000 book records and seven columns. This confirmed that the scraping process collected the intended number of records.

2. Average book price



The average listed price across the 1,000 books was **£35.07**.

This provided a general indication of the price level within the collected catalogue. However, the average alone does not describe how widely prices vary, so a price distribution chart and additional statistics were also useful for understanding the dataset.

3. Cheapest book

The least expensive book in the dataset was *An Abundance of Katherines*, priced at **£10.00**, with a five-star rating.

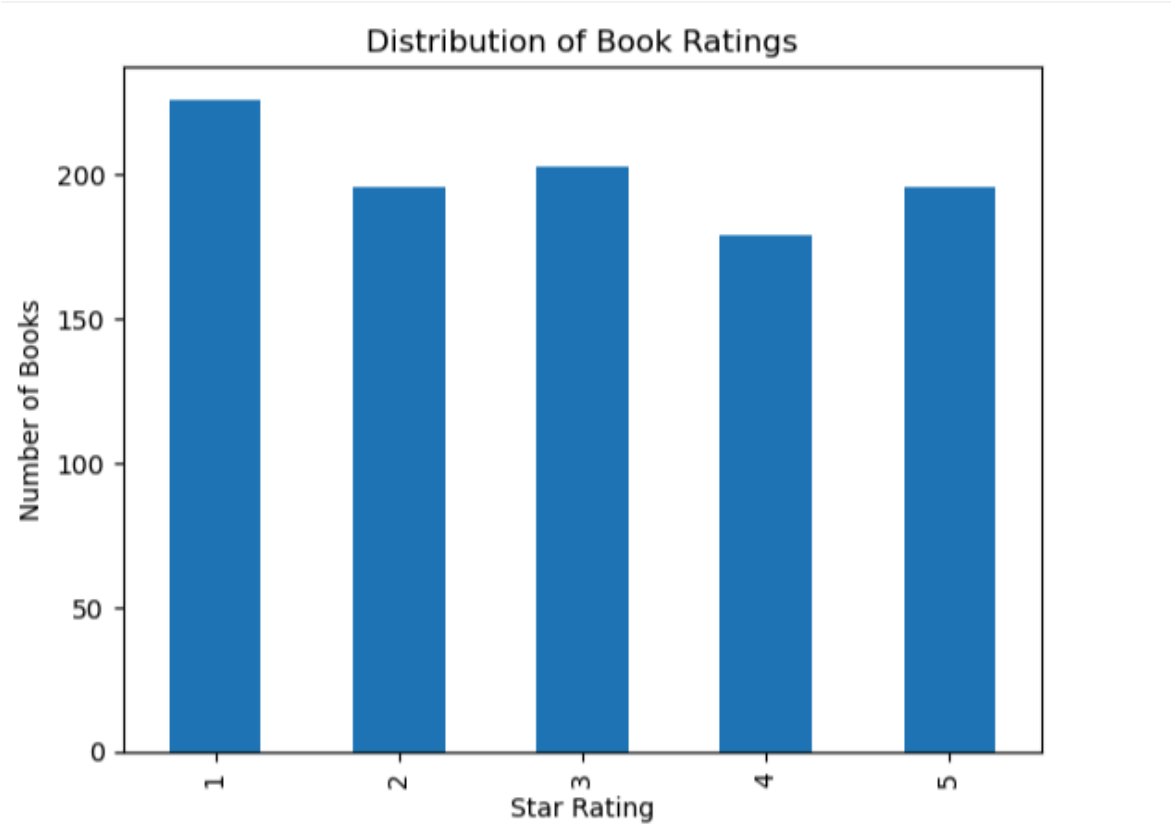
This finding demonstrates how sorting and filtering numerical values can be used to identify individual records of interest.

4. Most expensive book

The most expensive book was *The Perfect Play (Play by Play #1)*, priced at **£59.99**, with a three-star rating.

The difference between the lowest and highest prices shows that the catalogue contains books across a broad price range.

5. Distribution of book ratings



The rating analysis produced the following results:

Star rating	Number of books	Percentage
1 star	226	22.6%
2 stars	196	19.6%
3 stars	203	20.3%
4 stars	179	17.9%
5 stars	196	19.6%
Total	**1,000**	**100%**

One-star books formed the largest rating group, accounting for 22.6% of the dataset. Four-star books formed the smallest group, representing 17.9%.

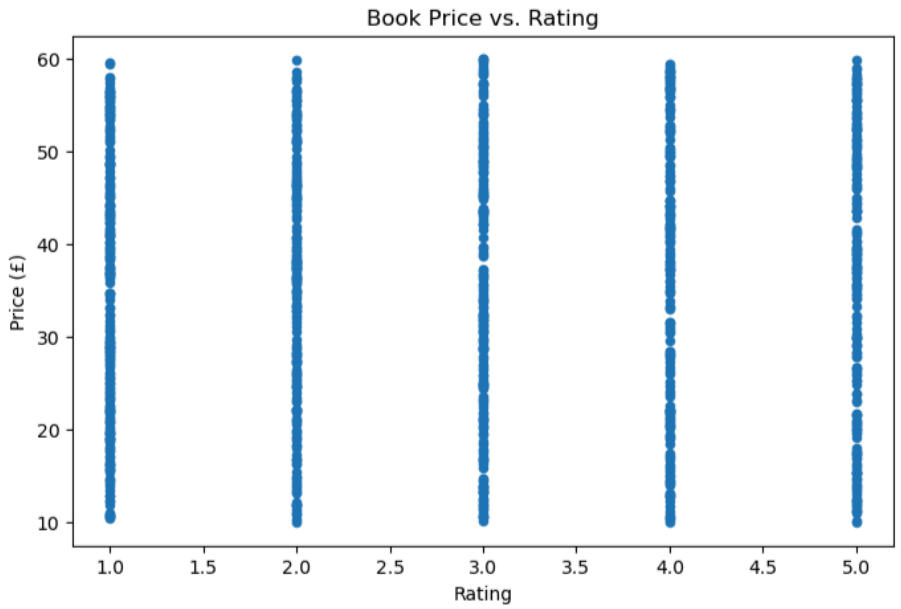
The distribution was relatively spread across the five rating categories, rather than being concentrated entirely in the highest or lowest rating.

6. Book availability

All 1,000 collected records were marked as *In stock*. The average price for this group was £35.07.

Because no other availability category appeared in the collected data, the analysis could not compare prices or stock levels between available and unavailable books. The finding describes the scraped catalogue at the time of collection and should not be interpreted as evidence that every book on the website is always in stock.

7. Relationship between price and rating



The correlation analysis compared the numerical price and rating columns. The calculated Pearson correlation coefficient was **0.03**.

This value indicates a negligible linear relationship between book price and star rating in the collected dataset. In practical terms, more expensive books did not consistently have higher ratings.

The grouped analysis supported this interpretation:

Star rating	Number of books	Average price (£)	Median price (£)
1 star	226	34.56	34.77
2 stars	196	34.81	36.22
3 stars	203	34.69	33.78
4 stars	179	36.09	37.80
5 stars	196	35.37	36.90

Four-star books had the highest average price at £36.09, while one-star books had the lowest average price at £34.56. The differences between the averages were relatively small compared with the overall price range.

These results suggest that price alone is not a reliable indicator of a book's star rating. However, the analysis establishes an association, not a causal relationship.

3.3 Visualizations Created in Python

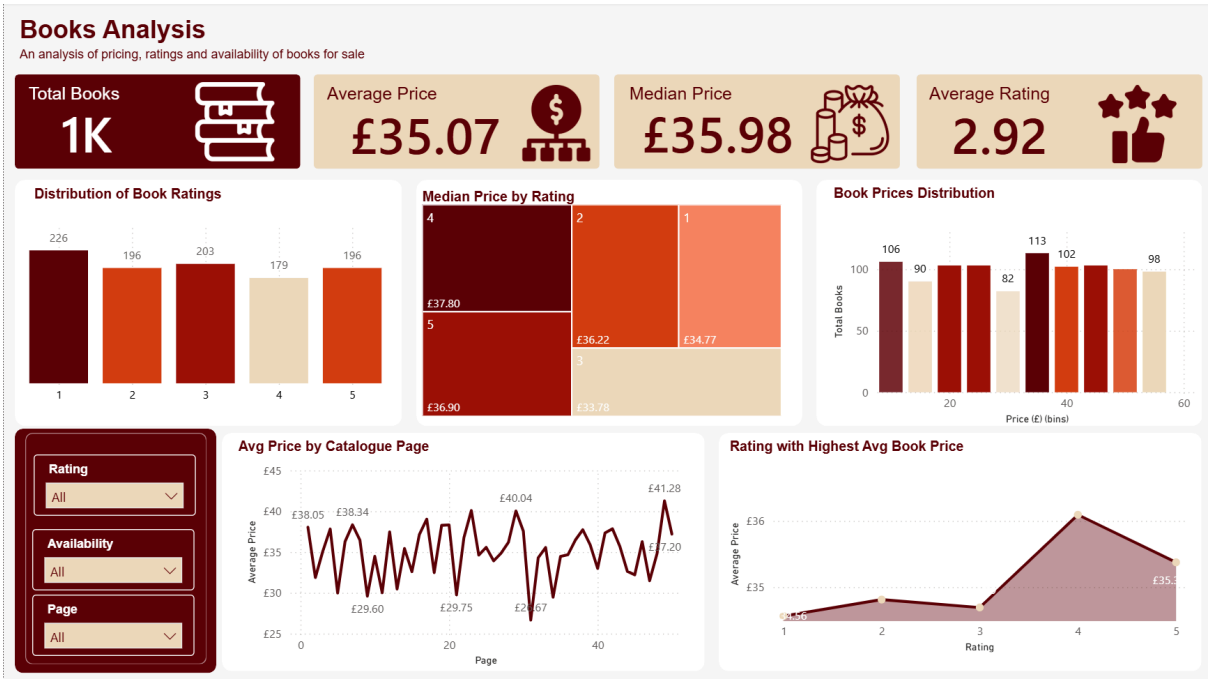
Several charts were developed to support the EDA findings:

- **Price distribution histogram:** Used to examine how book prices were distributed across the dataset.
- **Rating frequency bar chart:** Compared the number of books in each star-rating category.
- **Correlation matrix heatmap:** Displayed the correlation between price and rating.
- **Average price by rating bar chart:** Compared average prices across the five rating groups.

A grouped availability summary was also created to show the number of books and average price for each availability category.

Together, these outputs provided different views of the dataset and helped communicate the results more clearly than summary statistics alone.

4. Power BI Dashboard Analysis



4.1 Overview of the Dashboard

Following the exploratory data analysis conducted in Python, Power BI was used to develop an interactive dashboard titled **Books Analysis**. The dashboard provides a visual summary of book prices, ratings, and availability across the 1,000 records collected from Books to Scrape.

The purpose of the dashboard was to make the results of the Python analysis easier to interpret by presenting key performance indicators (KPIs), charts, and interactive filters in a single view.

The dashboard contains four KPI cards, four main analytical visualizations, a catalogue-page price trend, and slicers for filtering the data by rating, availability, and page.

4.2 Key Performance Indicators (KPIs)

The dashboard displays four headline metrics that summarize the collected dataset.

KPI	Result	Interpretation
Total Books	1,000	Represents the total number of book records collected.
Average Price	£35.07	Shows the mean listed price across the dataset.
Median Price	£35.98	Indicates the middle price when all book prices are arranged in ascending order.
Average Rating	2.92 out of 5	Shows the average numerical rating across the collected books.

The average price of £35.07 is close to the median price of £35.98. This suggests that the centre of the price distribution is relatively similar under these two measures. However, the difference alone is not sufficient to establish whether the distribution is symmetrical.

The average rating of 2.92 out of 5 indicates that the overall average rating is below three stars. This provides a useful summary of the catalogue's ratings, although it does not mean that every book received a low rating.

4.3 Distribution of Book Ratings

The *Distribution of Book Ratings* chart displays the number of books within each rating category.

Rating	Number of books	Percentage

1 star	226	22.6%
2 stars	196	19.6%
3 stars	203	20.3%
4 stars	179	17.9%
5 stars	196	19.6%

The chart shows that one-star books form the largest group, with 226 records, while four-star books form the smallest group, with 179 records.

The differences between the five categories are not particularly large. This indicates that the collected books are spread across the rating scale rather than being concentrated in only one category.

Observation: The presence of 226 one-star books suggests that lower-rated titles make up a substantial portion of the catalogue. However, the rating distribution alone cannot explain why these books received their ratings.

Recommendation: Book buyers could use the rating distribution as a starting point when comparing titles, but should also consider individual reviews, descriptions, and personal reading preferences before making purchasing decisions.

4.4 Median Price by Rating

The *Median Price by Rating* visualization compares the median selling price for books in each star-rating category.

Rating	Median price
1 star	£34.77
2 stars	£36.22

3 stars	£33.78
4 stars	£37.80
5 stars	£36.90

Four-star books have the highest median price at £37.80, while three-star books have the lowest median price at £33.78.

The results demonstrate that the typical price differs across rating categories. However, the pattern is not consistently increasing: five-star books have a lower median price than four-star books, while three-star books have the lowest median.

Observation: A higher rating does not automatically correspond to a higher median price. The four-star category has the highest median price, but the five-star category does not.

Recommendation: Buyers should not rely on price as a substitute for ratings or perceived quality. Similarly, sellers should avoid assuming that increasing a book's price will result in a higher customer rating.

4.5 Distribution of Book Prices

The *Book Prices Distribution* histogram illustrates how the 1,000 books are distributed across price intervals.

The visualization shows that books are spread across several price bands, with individual intervals containing different numbers of books. The dashboard also indicates that prices extend from the lower end of the catalogue to nearly £60.

Combined with the overall average price of £35.07 and median price of £35.98, the histogram provides a broader understanding of the catalogue's pricing structure.

Observation: The catalogue contains books at different price levels, meaning that customers can compare titles across a range of budgets. The concentration of books in particular price intervals may also help identify common pricing bands.

Recommendation: A seller or catalogue manager could use these price intervals to establish pricing categories, identify less populated price bands, and assess whether the catalogue offers a balanced range of price options.

4.6 Average Price by Catalogue Page

The *Avg Price by Catalogue Page* line chart compares the average price of books collected from each of the 50 catalogue pages.

The visualization shows noticeable fluctuations in average price across the pages. Some pages have average prices close to £30, while others reach above £40. The highest labelled value is approximately £41.28, while one of the lower labelled values is approximately £26.67.

This variation suggests that the books displayed on different pages do not all have the same average price.

However, the chart should be interpreted carefully. Catalogue pages are primarily a browsing and pagination structure, and the observed differences do not establish that the website deliberately groups books by price.

Observation: The average price varies from one catalogue page to another. This could reflect differences in the particular titles appearing on each page.

Recommendation: Further analysis should compare the price distribution and average price across pages to determine whether the differences are meaningful. If the catalogue is arranged according to another attribute, such as title or category, that information could help explain the observed pattern.

4.7 Rating with the Highest Average Book Price

The *Rating with Highest Avg Book Price* visualization compares average book prices across the five rating categories.

The results from the Python analysis indicate the following average prices:

Rating	Average price
1 star	£34.56
2 stars	£34.81
3 stars	£34.69
4 stars	£36.09
5 stars	£35.37

Four-star books have the highest average price at £36.09, followed by five-star books at £35.37. One-star books have the lowest average price at £34.56.

The difference between the highest and lowest category averages is £1.53. This is relatively small compared with the overall range of book prices.

The results reinforce the finding from the correlation analysis that price and rating have a negligible linear relationship, with a Pearson correlation coefficient of approximately 0.03.

Observation: Four-star books have the highest average price, but five-star books do not have the highest average price. This indicates that the relationship between price and rating is not consistently positive.

Recommendation: Customers should evaluate price and rating as separate decision factors. Additional variables, such as genre, publisher, book format, or publication information, could be investigated to determine whether they help explain differences in pricing.

4.8 Interactive Filters and Dashboard Usability

The dashboard includes three slicers: Rating, Availability, and Page. These controls allow users to filter the dataset and explore different subsets of books.

- **Rating slicer:** Allows users to examine prices and other measures for selected star ratings.
- **Availability slicer:** Allows users to investigate books according to their stock status.
- **Page slicer:** Allows users to examine records from specific catalogue pages.

These interactive controls improve the dashboard's usefulness because readers can move beyond the overall summaries and investigate particular segments of the dataset.

For example, a user could select five-star books and compare their prices with those of one-star books. Alternatively, selecting an individual catalogue page could help identify whether its average price differs from the overall catalogue average.

Since all 1,000 collected records were marked as in stock, the availability slicer currently has limited analytical value unless additional data with different stock statuses is collected.

5. Post-Analysis: Key Observations and Insights

The combined Python EDA and Power BI dashboard produced the following major observations.

Observation 1: The catalogue contains 1,000 book records

The scraping process collected 1,000 records across 50 catalogue pages. This provided a sufficiently large dataset for practising descriptive analysis, grouping, and visualization.

Insight: Automating the extraction process made it possible to collect and analyse a large number of records more efficiently than manually copying information from individual product pages.

Observation 2: Book prices are centred around £35

The average price is £35.07, while the median is £35.98. Both measures provide a useful summary of the catalogue's central pricing level.

Insight: The two measures are relatively close, but additional distribution statistics are needed to understand the spread and shape of the prices fully.

Observation 3: One-star books are the largest rating category

The one-star category contains 226 books, representing 22.6% of the dataset. Four-star books form the smallest category at 17.9%.

Insight: The rating distribution is relatively broad, with books represented across all five rating levels.

Observation 4: Higher prices do not necessarily mean higher ratings

Four-star books have the highest average price at £36.09, while five-star books average £35.37. The price-rating correlation is approximately 0.03.

Insight: There is almost no linear relationship between the two variables in the collected data. Other factors may influence pricing, but additional data would be needed to investigate them.

Observation 5: Average prices vary across catalogue pages

The dashboard shows fluctuations in average prices across the 50 pages, with some pages having noticeably higher average prices than others.

Insight: Page-level grouping provides another way to investigate the dataset. However, catalogue-page differences should not be treated as evidence of a deliberate pricing strategy without further investigation.

Observation 6: Availability analysis is limited by the collected data

All collected records were marked as in stock.

Insight: Although availability was captured successfully, the dataset does not contain enough variation in stock status to support comparisons between available and unavailable books.

6. Recommendations and Observations

Based on the findings from the Python EDA and Power BI dashboard, the following recommendations are proposed.

6.1 Recommendations for Further Data Analysis

1. Expand the dataset with additional book attributes.

Collect additional fields where available, such as book category, publisher, publication date, and product description. These attributes could help explain differences in price and ratings more effectively.

2. Examine ratings alongside other variables.

Since the relationship between price and rating is weak, future analysis should explore whether book category or other available attributes are associated with different rating patterns.

3. Improve stock availability analysis.

Collect availability information at different times or from a broader set of product listings. This would allow the analysis to investigate whether stock status changes and whether availability differs across product groups.

4. Validate data quality systematically.

Introduce checks for missing values, duplicate book records, invalid prices, unexpected rating values, and incomplete URLs. This would improve confidence in the results and make the scraping process more reliable.

7. Conclusion

This project demonstrated the practical application of web scraping, data cleaning, exploratory data analysis, and data visualization using Python and Power BI. Using Requests and BeautifulSoup, 1,000 book records were collected from Books to Scrape across 50 catalogue pages. The extracted data was cleaned, structured into a Pandas DataFrame, and exported to CSV and Excel formats for further analysis.

The Python EDA provided an initial understanding of the dataset, revealing an average book price of £35.07, a median price of £35.98, and an average rating of 2.92 out of 5. The analysis also identified *An Abundance of Katherines* as the least expensive book at £10.00 and *The Perfect Play (Play by Play #1)* as the most expensive at £59.99.

Building on these findings, the Power BI dashboard presented the results through interactive KPI cards, rating distributions, price comparisons, catalogue-page trends, and slicers for rating, availability, and page. The visual analysis confirmed that one-star books formed the largest rating category, while four-star books had the highest average and median prices. It also revealed variations in average prices across catalogue pages, although further investigation would be required to explain these differences.

Overall, the project demonstrated how Python and Power BI can be combined to transform scraped website information into structured data, meaningful insights, and an interactive reporting solution. It strengthened practical skills in web scraping, data transformation, statistical analysis, and dashboard development while highlighting the importance of data quality and careful interpretation. Future improvements could include additional data validation, collecting more book attributes, and expanding the analysis to investigate other factors that may influence book prices and ratings.